Truncated Whitehead principle, base case: a surjective-on-truncations and fully faithful map is an equivalence

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:whitehead0' (source lines 2378-2385, Theorem 8.8.1); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own introduction to the downward induction that proves the truncated Whitehead principle, which is what makes the role of this theorem visible.

Editorial note. The retained counter values (chapter 8, section 8, theorem offset 0) make the retained theorem print as Theorem 8.8.1, matching the original book numbering.

Editorial note. The two external references point at the section on monomorphisms and surjections of chapter 4 and at the characterisation of equivalences recorded there; the counted proof is retained in full.

Editorial scope. One original result of the HoTT Book, counted once: the base case of the truncated Whitehead principle (Theorem 8.8.1).

Allowed standard prerequisites (editorial).

- 0-truncation. trunc 0 A is the set of connected components of A, and a map is surjective on 0-truncations when every component of the codomain is hit.. It is the first hypothesis of the statement and is used at source line 2388 to produce a point a : A.
- Embedding. f is an embedding when ap f : (x = y) -> (f x = f y) is an equivalence for all x, y.. The proof observes at source line 2387 that the second hypothesis says exactly that f is an embedding.
- Characterisation of equivalences by embeddings that are surjective. A map is an equivalence if and only if it is an embedding and surjective.. It is the reduction performed at source line 2388 and it is the reason the rest of the proof only has to produce surjectivity.

Source reference. Original theorem 4.6.3: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1824–1824. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the

counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original section 4.6: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1680–1680. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

It is beyond the scope of this book to describe any models of type theory, so we will not explain how Whitehead's principle might fail in some of them. However, we can prove that it holds whenever the types involved are n-truncated for some finite n, by "downward" induction on n. In addition to being of interest in its own right (for instance, it implies the essential uniqueness of Eilenberg–Mac Lane spaces), the proof of this result will hopefully provide some intuitive explanation for why we cannot hope to prove an analogous theorem without truncation hypotheses.

We begin with the following modification of original theorem 4.6.3, which will eventually supply the induction step in the proof of the truncated Whitehead's principle. It may be regarded as a type-theoretic, ∞-groupoidal version of the classical statement that a fully faithful and essentially surjective functor is an equivalence of categories.

Theorem 8.8.1. *Suppose $f : A \to B$ is a function such that*

(i) $\|f\|_0 : \|A\|_0 \to \|B\|_0$ is surjective, and
(ii) for any $x, y : A$, the function $\mathsf{ap}_f : (x =_A y) \to (f(x) =_B f(y))$ is an equivalence.

Then f is an equivalence.

Proof. Note that (ii) is precisely the statement that f is an embedding, c.f. original section 4.6. Thus, by original theorem 4.6.3, it suffices to show that f is surjective, i.e. that for any $b : B$ we have $\|\mathsf{fib}_f(b)\|_{-1}$. Suppose given b; then since $\|f\|_0$ is surjective, there merely exists an $a : A$ such that $\|f\|_0(|a|_0) = |b|_0$. And since our goal is a mere proposition, we may assume given such an a. Then we have $|f(a)|_0 = \|f\|_0(|a|_0) = |b|_0$, hence $\|f(a) = b\|_{-1}$. Again, since our goal is still a mere proposition, we may assume $f(a) = b$. Hence $\mathsf{fib}_f(b)$ is inhabited, and thus merely inhabited. □